

Geonwoo Kim

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EDUCATION

- University of Chicago**2025–Present
- Graduate Student at Large, Department of Economics*
PhD coursework: ECON 31000 Empirical Analysis I (in progress).
- Grinnell College**2019–2024
- B.A. in Mathematics with Honors; Concentration in Statistics.*
Selected coursework: Real Analysis, Numerical Analysis, Mathematical Statistics, Advanced Econometrics, Measure Theory, Measure-Theoretic Probability, Empirical Process Theory.

FIELDS

- Econometric theory** High-dimensional inference, measurement error, computational econometrics.
Applied econometrics Policy evaluation, labor economics, economics of AI.

RESEARCH

1. *Denoised Partial Correlation for Conditional Independence Tests under Noisy Confounders* (with Suyong Song and Jonathan Wells). Under revision.
2. *Double/Debiased CoCoLasso of Treatment Effects with Mismeasured High-Dimensional Control Variables* (with Suyong Song). Accepted at Midwest Econometrics Group 2024; under revision.
3. *Signal Space Conjugate Gradient Iterative Hard Thresholding* (with Kate Bartz and Jeffrey Blanchard). Under review.
4. *Inference After AI Measurement* in progress.

EXPERIENCE

- Research Professional in Econometric Theory**2025.07–Present
- University of Chicago, Department of Economics*
- Inference for Linear Systems with Unknown Coefficients**
- PIs: Yuehao Bai, Kirill Ponomarev, Andres Santos, Azeem Shaikh, Max Tabord-Meehan, Alexander Torgovitsky*
- Work on tests for whether a data-generating process lies in sets defined by large systems of linear restrictions, with applications to nonparametric IV and shape restrictions.
 - Contribute to uniform Gaussian and bootstrap approximations for supremum-type statistics; implement procedures in R, solve linear and mixed-integer programs, and run parallelized Monte Carlo experiments.
- The Identifying Power of an Assumption: Comparing Sharp Identified Sets under Linear Restrictions**
- PIs: Yuehao Bai, Shunzhuang Huang, Azeem Shaikh, Edward Vytlačil*
- Audit and extend Python code for a computational partial-identification project comparing sharp identified sets under alternative linear restrictions in finite-dimensional models.
 - Replace sampled or heuristic dual-piece discovery with exact polyhedral enumeration using cdd/pycddlib; verify endpoints, redundancy diagnostics, pruning routines, ordinal-IV computations, and replication-package output.
- Inference for Policy Learning and Optimal Welfare**
- PIs: Kirill Ponomarev and Vira Semenova*
- Assist with computational and theoretical components of a policy-learning project on welfare bounds and confidence bands for data-driven treatment rules.
 - Implement simulation and replication code, check analytical derivations, and diagnose finite-sample performance of proposed inference procedures.
- Undergraduate Researcher**2024.08–2024.12
- Grinnell College, Department of Statistics*
Advisor: Jonathan Wells
- Used random matrix theory and representation theory to develop an efficient statistic for a permutation-based conditional-independence test under noisy confounders.
 - Formulated an empirical risk minimization procedure that minimizes rank mismatch in estimated residuals; implemented the algorithm in R and ran simulations.

	Undergraduate Researcher <i>University of Iowa, Department of Economics</i> <i>Advisor: Suyong Song</i>	2023.08–Present
	- Developed a high-dimensional errors-in-variables regression procedure based on an orthogonal score that mitigates measurement error and regularization bias in nuisance estimation. - Constructed a measurement-error covariance estimator using auxiliary information and the empirical spectral distribution; derived the influence function for asymptotic variance and uniform inference. - Ran large-scale Monte Carlo simulations showing improved coverage and reduced omitted-variable bias relative to naive estimators.	
	Undergraduate Researcher <i>Grinnell College, Department of Mathematics</i> <i>Advisor: Jeffrey Blanchard</i>	2023.05–Present
	- Developed Signal Space Conjugate Gradient Iterative Hard Thresholding, a sparse optimization algorithm for dictionary-domain signal recovery. - Proved convergence and noise-stability results under bounded noise and alternative dictionary choices; implemented and benchmarked SSCGIHT in Python.	
	Research Assistant <i>Seoul National University, Department of Operations Research</i> <i>PI: Song-Hee Kim</i>	2022.03–2024.03
	- Studied the causal impact of a shift-choice system on nurse turnover and service quality using administrative shift rosters, turnover records, and quality indicators from a large hospital. - Constructed a nurse-month panel; inferred department-level treatment timing under staggered adoption; assisted with generalized DiD, event-study, robustness, and sensitivity analyses.	
	Research Assistant <i>Science and Technology Policy Institute, South Korea</i> <i>PI: Minhyung Lee</i>	2019.02–2019.06
	Conducted policy research on cross-institutional public R&D collaboration networks; implemented graph-based algorithms to identify mediator institutions in innovation diffusion.	
TEACHING	<i>Grinnell College</i>	
	STA 336: Probability and Statistics I, Course Mentor	Fall 2024
	ECN 280: Microeconomic Analysis, Course Grader	Fall 2024
	STA 295: Statistical Learning, Course Mentor	Spring 2024
	ECN 286: Econometrics, Course Mentor	Fall 2023, Spring 2024
	<i>Hosanna International School, Busan, South Korea</i>	
	High School Calculus, Teaching Assistant	2020.05–2020.08
SERVICE	Korean Students Association President, Grinnell College	2022–2023
	Squad Leader, Republic of Korea Army	2021–2022
	Drill Sergeant, Republic of Korea Army	2020–2022
SKILLS	Programming Python, R, MATLAB, Stata/Mata, LaTeX, Git.	
	Languages Korean (native), English (fluent), German (intermediate).	